

LAPORAN PRAKTIKUM – BUBBLE SORT

Laporan ini disusun untuk memenuhi Tugas ke-7 Praktikum Algoritma & Pemrograman

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SOAL BUBBLE SORT_07 :

Berdasarkan contoh no1, urutkan secara Descending tanpa inputan (nama file : PASD7-07.ccp):

SOURCE CODE:

```
#include <iostream>
```

```
using namespace std;
```

```
int data[10] = {12, 5, 9, 20, 15, 7};
```

```
int n = 6;
```

```
void tukar(int a, int b)
```

```
{
```

```
    int t;
```

```
    t = data[a];
```

```
    data[a] = data[b];
```

```
    data[b] = t;
```

```
}
```

```
void tampil()
```

```
{
```

```
    for(int i = 0; i < n; i++)
```

```
    {
```

```
        cout << data[i] << " ";
```

```
    }
```

```
    cout << endl;
```

```
}
```

```
void bubble_sort_descending()
```

```
{
```

```
    for(int i = 1; i < n; i++)
```

```
    {
```

```
        for(int j = n - 1; j >= i; j--)
```

```
        {
```

```
            if(data[j] > data[j - 1])
```

```
            {
```

```
                tukar(j, j - 1);
```

```
            }
```

```
        }
```

```
        tampil();
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
    cout << "===== " << endl;
```

```
    cout << "    BUBBLE SORT DESCENDING    " << endl;
```

```
    cout << "===== " << endl;
```

```
    cout << "Data sebelum diurutkan : " << endl;
```

```
    tampil();
```

```
cout << endl;
```

```
cout << "Proses Bubble Sort Descending : " << endl;
```

```
bubble_sort_descending();
```

```
cout << endl;
```

```
cout << "Data setelah diurutkan : " << endl;
```

```
tampil();
```

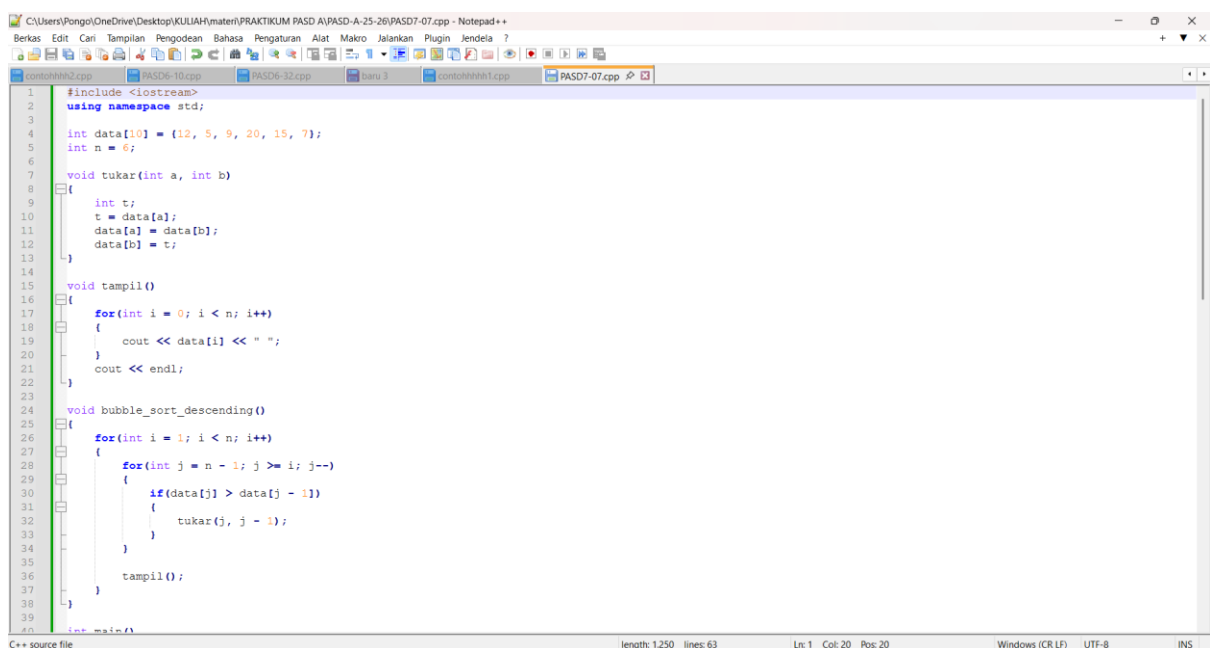
```
cout << "===== " << endl;
```

```
cout << "    TERIMA KASIH    " << endl;
```

```
cout << "===== " << endl;
```

```
return 0;
```

```
}
```



```
1 #include <iostream>
2 using namespace std;
3
4 int data[10] = {12, 5, 9, 20, 15, 7};
5 int n = 6;
6
7 void tukar(int a, int b)
8 {
9     int t;
10    t = data[a];
11    data[a] = data[b];
12    data[b] = t;
13 }
14
15 void tampil()
16 {
17     for(int i = 0; i < n; i++)
18     {
19         cout << data[i] << " ";
20     }
21     cout << endl;
22 }
23
24 void bubble_sort_descending()
25 {
26     for(int i = 1; i < n; i++)
27     {
28         for(int j = n - 1; j >= i; j--)
29         {
30             if(data[j] > data[j - 1])
31             {
32                 tukar(j, j - 1);
33             }
34         }
35         tampil();
36     }
37 }
38
39 int main()
40 {
```

C++ source file length: 1250 lines: 63 Ln: 1 Col: 20 Pos: 20 Windows (CR LF) UTF-8 INS

```
C:\Users\Pongo\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26\PASD7-07.cpp - Notepad++
Berkas Edit Cari Tampilan Pengodean Bahasa Pengaturan Alat Makro Jalankan Plugin Jendela ?
PASD6-10.cpp PASD6-32.cpp baru 3 console1.cpp PASD7-07.cpp
31 {
32     tukar(j, j - 1);
33 }
34 }
35 tampil();
36 }
37 }
38 }
39
40 int main()
41 {
42     cout << "===== " << endl;
43     cout << "      BUBBLE SORT DESCENDING      " << endl;
44     cout << "===== " << endl;
45
46     cout << "Data sebelum diurutkan : " << endl;
47     tampil();
48
49     cout << endl;
50     cout << "Proses Bubble Sort Descending : " << endl;
51
52     bubble_sort_descending();
53
54     cout << endl;
55     cout << "Data setelah diurutkan : " << endl;
56     tampil();
57
58     cout << "===== " << endl;
59     cout << "      TERIMA KASIH      " << endl;
60     cout << "===== " << endl;
61
62     return 0;
63 }
```

C++ source file | length: 1.250 | lines: 63 | Ln: 1 | Col: 20 | Pos: 20 | Windows (CR LF) | UTF-8 | INS

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Pongo\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26>g++ -o m PASD7-07.cpp

C:\Users\Pongo\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26>m
=====
      BUBBLE SORT DESCENDING
=====
Data sebelum diurutkan :
12 5 9 20 15 7

Proses Bubble Sort Descending :
20 12 5 9 15 7
20 15 12 5 9 7
20 15 12 9 5 7
20 15 12 9 7 5
20 15 12 9 7 5

Data setelah diurutkan :
20 15 12 9 7 5
=====
      TERIMA KASIH
=====

C:\Users\Pongo\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26>|
```

SOAL BUBBLE SORT_10:

Berdasarkan contoh no1, buat dengan menggunakan struktur (struct), tanpa inputan dan melakukan pengurutan secara Ascending (nama file : PASD7 10.ccp):

SOURCE CODE:

```
#include <iostream>
```

```
using namespace std;
```

```
struct Data
```

```
{
```

```
    int angka[10];
```

```
    int n;
```

```
};
```

```
Data nilai = {{25, 12, 9, 30, 18, 5}, 6};
```

```
void tukar(int &a, int &b)
```

```
{
```

```
    int t;
```

```
    t = a;
```

```
    a = b;
```

```
    b = t;
```

```
}
```

```
void tampil()
```

```
{
```

```
    for(int i = 0; i < nilai.n; i++)
```

```
{  
    cout << nilai.angka[i] << " ";  
}  
  
cout << endl;  
}  
  
void bubble_sort()  
{  
    for(int i = 1; i < nilai.n; i++)  
    {  
        for(int j = nilai.n - 1; j >= i; j--)  
        {  
            if(nilai.angka[j] < nilai.angka[j - 1])  
            {  
                tukar(nilai.angka[j], nilai.angka[j - 1]);  
            }  
        }  
    }  
  
    tampil();  
}  
  
cout << endl;  
}
```

```
int main()

{

    cout << "===== " << endl;

    cout << "  BUBBLE SORT ASCENDING STRUCT  " << endl;

    cout << "===== " << endl;


    cout << "Data sebelum diurutkan : " << endl;

    tampil();


    cout << endl;

    cout << "Proses Bubble Sort Ascending : " << endl;


    bubble_sort();


    cout << "Data setelah diurutkan : " << endl;

    tampil();


    cout << "===== " << endl;

    cout << "      TERIMA KASIH      " << endl;

    cout << "===== " << endl;


    return 0;

}
```



```
C:\Users\Pongol\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26\PASD7-10.cpp - Notepad++
Berkas Edit Cari Tampilan Pengodean Bahasa Pengaturan Alat Makro Jalankan Plugin Jendela ?
contoh1.cpp PASD6-10.cpp PASD6-32.cpp baru 3 contoh1.cpp PASD7-07.cpp PASD7-10.cpp
1 #include <iostream>
2 using namespace std;
3
4 struct Data
5 {
6     int angka[10];
7     int n;
8 };
9
10 Data nilai = {{25, 12, 9, 30, 18, 5}, 6};
11
12 void tukar(int &a, int &b)
13 {
14     int t;
15     t = a;
16     a = b;
17     b = t;
18 }
19
20 void tampil()
21 {
22     for(int i = 0; i < nilai.n; i++)
23     {
24         cout << nilai.angka[i] << " ";
25     }
26     cout << endl;
27 }
28
29 void bubble_sort()
30 {
31     for(int i = 1; i < nilai.n; i++)
32     {
33         for(int j = nilai.n - 1; j >= i; j--)
34         {
35             if(nilai.angka[j] < nilai.angka[j - 1])
36             {
37                 tukar(nilai.angka[j], nilai.angka[j - 1]);
38             }
39         }
40     }
41 }
```

C++ source file length: 1329 lines: 70 Ln: 1 Col: 20 Pos: 20 Windows (CR LF) UTF-8 INS

```
C:\Users\Pongol\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26\PASD7-10.cpp - Notepad++
Berkas Edit Cari Tampilan Pengodean Bahasa Pengaturan Alat Makro Jalankan Plugin Jendela ?
contoh1.cpp PASD6-10.cpp PASD6-32.cpp baru 3 contoh1.cpp PASD7-07.cpp PASD7-10.cpp
34 for(int j = nilai.n - 1; j >= i; j--)
35 {
36     if(nilai.angka[j] < nilai.angka[j - 1])
37     {
38         tukar(nilai.angka[j], nilai.angka[j - 1]);
39     }
40 }
41
42 tampil();
43
44 cout << endl;
45 }
46
47 int main()
48 {
49     cout << "===== " << endl;
50     cout << " BUBBLE SORT ASCENDING STRUCT " << endl;
51     cout << "===== " << endl;
52
53     cout << "Data sebelum diurutkan : " << endl;
54     tampil();
55
56     cout << endl;
57     cout << "Proses Bubble Sort Ascending : " << endl;
58
59     bubble_sort();
60
61     cout << "Data setelah diurutkan : " << endl;
62     tampil();
63
64     cout << "===== " << endl;
65     cout << " TERIMA KASIH " << endl;
66     cout << "===== " << endl;
67
68     return 0;
69 }
70
```

C++ source file length: 1329 lines: 70 Ln: 1 Col: 20 Pos: 20 Windows (CR LF) UTF-8 INS

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8457]
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C:\Users\Pongo\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26>g++ -o i PASD7-10.cpp

C:\Users\Pongo\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26>i
=====
      BUBBLE SORT ASCENDING STRUCT
=====
Data sebelum diurutkan :
25 12 9 30 18 5

Proses Bubble Sort Ascending :
5 25 12 9 30 18
5 9 25 12 18 30
5 9 12 25 18 30
5 9 12 18 25 30
5 9 12 18 25 30

Data setelah diurutkan :
5 9 12 18 25 30
=====
          TERIMA KASIH
=====

C:\Users\Pongo\OneDrive\Desktop\KULIAH\materi\PRAKTIKUM PASD A\PASD-A-25-26>
```